

HealthTech



Project Name: Patient Assistance Alert System



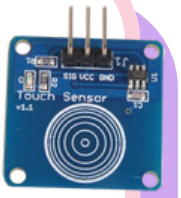
The Problem: A patient is resting alone in a hospital room and suddenly needs help from the nurse. The nurse is busy helping other patients and may not notice the patient's need right away. The patient cannot easily leave the bed to call for help. A student thinks there should be a simple way for the patient to call the nurse.

🎯 **What You Will Design:** Students will build a smart Patient Assistance Alert System that allows patients to request help by touching a sensor. The system will activate a buzzer, turn on an LED, and display an alert message on the OLED screen.

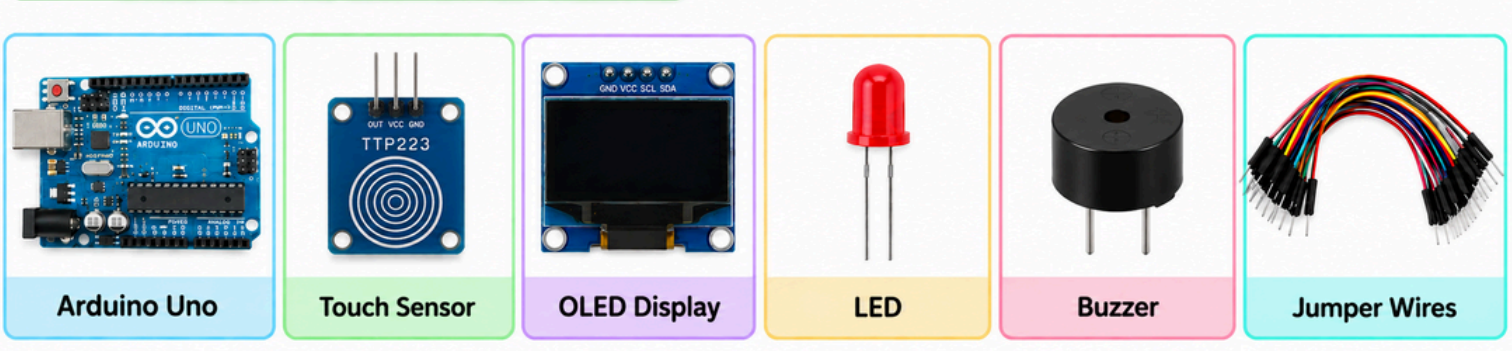


👆 What is a touch sensor?

A touch sensor can detect when a finger touches its surface. It works like a touch button and sends a signal to the Arduino when someone touches it.
Example: 📱 The touchscreen on a smartphone responds when you touch it. A touch sensor works in a similar way to detect human touch.



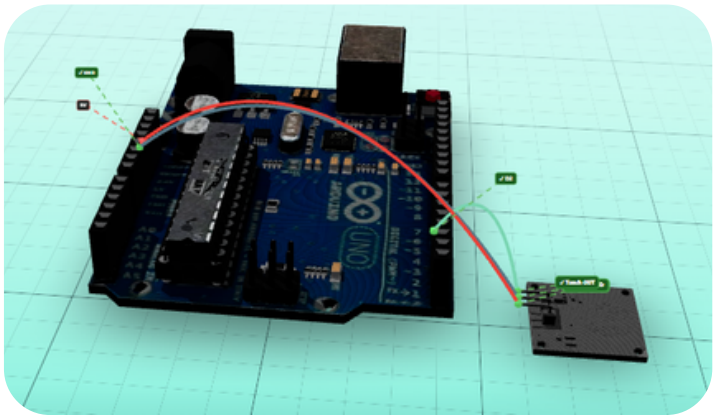
Components Required



Circuit Designing

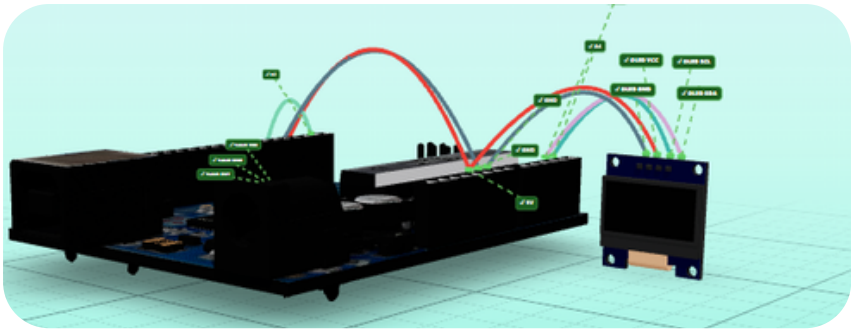
Step 1: Connect the touch sensor to the Arduino Uno.

Sensor Pins	Arduino Pins
Vcc pin	5V Pin
GND Pins	GND Pin
OUT Pin	Digital Pin 2



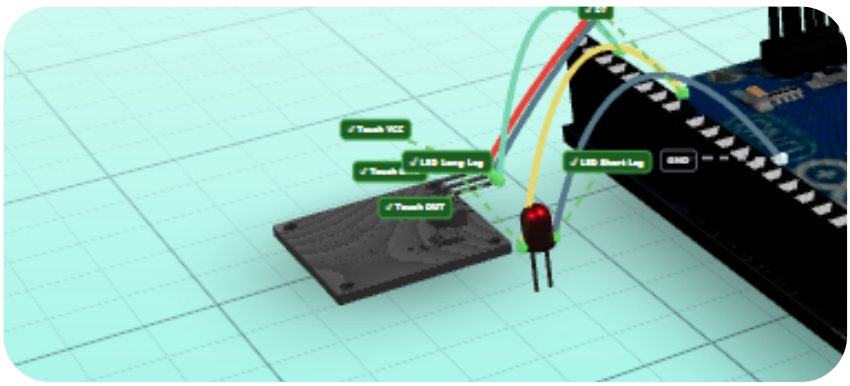
Step 2: Connect the OLED display to the Arduino Uno.

OLED Pins	Arduino Pins
Vcc pin	5V Pin
GND Pins	GND Pin
SDA Pin	Analog A4 Pin
SCL Pin	Analog A5 Pin



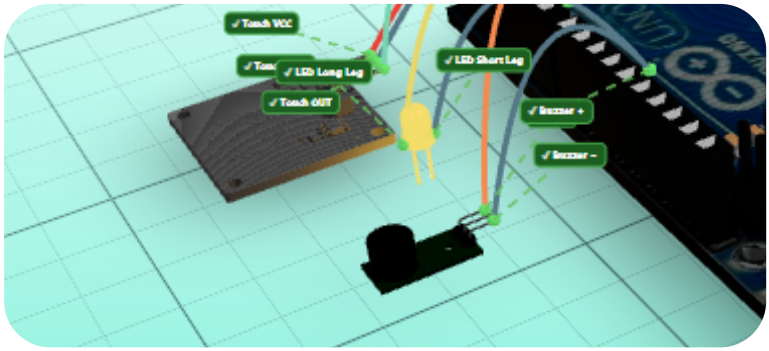
Step 3: Connect the LED to the Arduino Uno.

LED Pins	Arduino Uno
Long Leg	Digital Pin 7
Short Leg	GND Pin



Step 4: Connect the buzzer to the Arduino Uno.

Buzzer Pins	Arduino Uno
Positive Pin	Digital Pin 8
Negative Pin	GND Pin



Project Coding

1. Open the Arduino software and add the OLED libraries and define the sensor pins.

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

// Pin definitions
#define TOUCH_SENSOR_PIN 2
#define LED_PIN 7
#define BUZZER_PIN 8
```

2. Make code to make the OLED settings.

```
// OLED display settings
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);
```

3. Make code to initialize the serial communication and the sensor pins.

```
void setup() {
  // Initialize serial communication
  Serial.begin(9600);

  // Initialize pins
  pinMode(TOUCH_SENSOR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);
```

4. Make code to initialize the OLED display and print the patient safe message.

```
  // Initialize OLED display
  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println(F("SSD1306 allocation failed"));
    for (;;);
  }
  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(SSD1306_WHITE);
  display.setCursor(0, 0);
  display.println("Patient Safe");
  display.display();
}
```

5. Make code to check if the touch sensor is activate then turn on the LED and the buzzer.

```
void loop() {
  // Check if the touch sensor is activated
  if (digitalRead(TOUCH_SENSOR_PIN) == HIGH) {
    // Turn on LED and buzzer
    digitalWrite(LED_PIN, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);
    // Update OLED display
    display.clearDisplay();
    display.setCursor(0, 0);
    display.println("Assistance Required");
    display.display();
```

6. Make code to turn off the LED and the buzzer.

```
  } else {
    // Turn off LED and buzzer
    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
```

7. Make code to display the 'patient safe' message on the OLED display.

```
    // Update OLED display
    display.clearDisplay();
    display.setCursor(0, 0);
    display.println("Patient Safe");
    display.display();
  }

  delay(100); // Small delay to debounce the touch sensor
}
```

Project Output



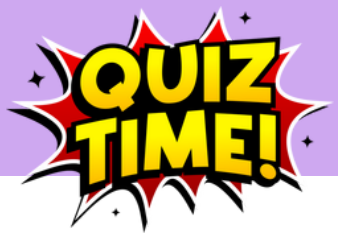


🌾 Where Is This Used in Real Life?

Similar patient assistance systems are used in hospitals, nursing homes, and healthcare centers. Patients can quickly request help without leaving their beds, allowing nurses and caregivers to respond faster.

Learning Outcomes

- The students will be able to use a touch sensor to detect user input.
- The students will be able to write code that activates alerts based on sensor inputs.
- The students will be able to build a smart patient-assistance system using sensors and actuators.



Section 1: Multiple Choice Questions

1. What problem does the student want to solve?

a) The patient needs a bigger bed
b) The patient may need help when no one is nearby
c) The hospital needs more rooms
d) The patient needs a new phone

2. Which component does the patient touch to ask for help?

a) OLED Display
b) Buzzer
c) Touch Sensor
d) LED

3. What happens when the patient touches the sensor?

a) The system activates an alert
b) The Arduino turns off
c) The OLED stops working
d) The sensor disconnects

Section 2: True and False Statements

1. A patient may sometimes need help when no one is nearby.....

2. The buzzer makes a sound to alert others when help is needed.....

3. The touch sensor is used to display messages on the OLED screen.....